

FLOODWATCH

Complete IoT, AI, Machine Learning and Real-Time Flood Monitoring Prototype Documentation

AI-Assisted IoT-Based Urban Flood Monitoring and Early Warning System — Cyber-Physical Engineering Blueprint

Authoritative System Architecture, Forensic Audit, Experimental ML Analysis & Physical Prototype Construction Blueprint

This master technical document specifies the complete FloodWatch cyber-physical platform from inception to production verification. It details the verified software stack (FastAPI, PostgreSQL 16.1, Alembic, React 18, WebSocket dispatcher, and 9-model XGBoost quantile inference suite) and provides an exhaustive engineering roadmap for constructing, wiring, calibrating, and demonstrating the benchtop physical prototype using an ESP32 microcontroller, waterproof ultrasonic sensor, capacitive soil moisture probe, tipping bucket rain gauge, and MPU-6050 accelerometer.

Prepared for: B.Tech Major Project Final Documentation, College Academic Evaluation, Technical Viva Voce Defense, and Future Field Deployment.

Document Title:	FloodWatch — Complete IoT, AI, Machine Learning and Real-Time Flood Monitoring Prototype Documentation
Document Classification:	B.Tech Final Project Report / Engineering Specification / Viva Preparation Guide
Active Software Stack:	FastAPI (Python 3.13) + PostgreSQL 16.1 + React 18 / Vite / TypeScript + WebSockets
Safety Governance:	Deterministic 5-Factor Rule-Based Hydrological Engine (Sovereign Primary Safety Authority)
Machine Learning:	Direct Multi-Horizon XGBoost (+1h, +3h, +6h) with Dedicated Q05/Q95 Pinball Loss Corridors
Physical Prototype:	ESP32 DevKit v1 + JSN-SR04T Waterproof Ultrasonic + Capacitive Soil + Rain Gauge + MPU6050
Inspiration Catchment:	Ertf River Basin, Bad Münstereifel, North Rhine-Westphalia, Germany (2021-2023)
Status Designations:	[VERIFIED IMPLEMENTATION] [EXPERIMENTAL ML] [PHYSICAL PROTOTYPE PLAN]

PRIMARY SAFETY GOVERNANCE INVARIANT

MANDATORY SAFETY ARCHITECTURE INVARIANT:

The deterministic Rule-Based Hydrological Risk Engine is the **SOVEREIGN PRIMARY SAFETY AUTHORITY**. Machine Learning predictions are strictly advisory decision-support signals and NEVER possess the authority to override, downgrade, delay, or suppress deterministic safety alarms or physical threshold breaches. Even if ML inference fails, server connectivity drops, or models project receding water, the deterministic threshold-based safety system operates autonomously and inviolably.

SECTION 1 — PROJECT EXECUTIVE OVERVIEW

1.1 What FloodWatch Is: FloodWatch is an end-to-end cyber-physical environmental monitoring and early warning system designed to safeguard communities situated in flood-vulnerable river basins and urban catchments. It integrates physical edge sensing, high-throughput REST telemetry ingestion, persistent relational database storage, a deterministic 5-factor hydrological safety engine, multi-horizon machine learning forecasting, and an interactive, real-time web dashboard.

1.2 The Problem of Urban and River Flooding: River valleys with steep side-slopes and dense urban centers face catastrophic rapid-onset flash floods. When intense convective storm clusters stall over small catchments, the catchment response time (time of concentration, T_c) is exceedingly short (often less than 60 minutes). Impervious urban surfaces and saturated soils prevent infiltration, turning gentle streams into violent torrents that destroy bridges, submerge roads, and trap residents before regional meteorological agencies can issue warnings.

1.3 Why Real-Time Flood Monitoring is Critical: Traditional regional weather forecasts rely on satellite radar and macro-scale hydrological models running on 6- to 12-hour batch cycles. These systems cannot resolve hyper-local flash floods. A sub-second in-situ monitoring network provides the 30 to 60 minutes of actionable lead time needed to automate floodgates, deploy sirens, sound emergency cell broadcasts, and evacuate ground-floor structures.

1.4 The Complete FloodWatch Solution: FloodWatch unites edge IoT nodes, cloud microservices, and AI to deliver early warnings. Edge transducers continuously capture water level, water rise rate (dh/dt), rainfall intensity, soil moisture, and pole tilt. Data is pushed over Wi-Fi/HTTP to an asynchronous FastAPI backend, validated and stored in PostgreSQL 16, analyzed by a deterministic risk engine, projected forward by 9 XGBoost models (+1h, +3h, +6h), and pushed to browser operators in under 1.5 milliseconds via WebSockets.

1.5 Project Objectives:

- Engineer an affordable, open-hardware IoT river station using the dual-core ESP32 and industrial-grade waterproof sensors.
- Establish an asynchronous backend capable of high-frequency telemetry ingestion with cryptographic API key authentication.
- Enforce an inviolable safety hierarchy where physical threshold breaches trigger immediate alarms regardless of ML state.
- Implement direct multi-horizon ML forecasting with dedicated pinball loss quantile uncertainty corridors rather than arbitrary heuristics.
- Build a responsive operator dashboard featuring live GIS mapping, interactive Chart.js time-series, and scenario simulators.

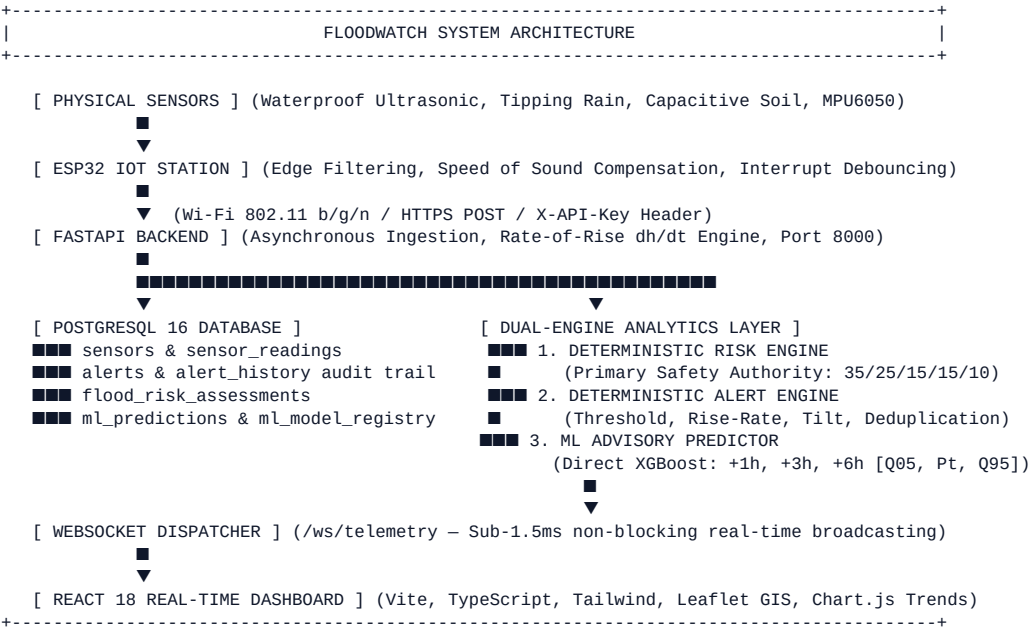
1.6 What Makes FloodWatch Different from a Normal Dashboard: Conventional IoT dashboards are passive visualizers that display static historical data from a database. FloodWatch is an active decision-support and life-safety platform featuring:

1. *Rate-of-Rise (dh/dt) Velocity Detection:* Flags rapidly rising waters before absolute warning thresholds are physically breached.
2. *Dual-Engine Architecture:* Strict segregation of deterministic safety rules (primary) from experimental machine learning (advisory).
3. *Statistically Valid Quantile Prediction:* Uses dedicated pinball-loss models for Q05 and Q95 bounds, eliminating heuristic multipliers.
4. *Cyber-Physical Synergy:* Seamlessly accepts live hardware telemetry or calibrated hydrodynamic scenario simulations.

1.7 How IoT + Backend + Database + Risk Engine + ML + Dashboard Work Together:

1.7 High-Level Cyber-Physical System Architecture

The diagram below details the end-to-end data flow connecting the physical sensing station, asynchronous FastAPI ingestion, relational PostgreSQL persistence, dual analytics engines, WebSocket broadcasting, and the React operator dashboard:



SECTION 2 — CURRENT PROJECT IMPLEMENTATION STATUS

Every subsystem in the FloodWatch project was audited directly against the production codebase, PostgreSQL database, and firmware. To maintain absolute academic and scientific integrity, completed software components are strictly distinguished from upcoming physical hardware implementations.

Subsystem	Technology Stack	Verified Architecture & Features	Implementation Status
Frontend UI	React 18, TypeScript, Vite, Tailwind CSS, Leaflet, Chart.js	11 modular components: Header, Sidebar, MetricCards, SensorMap, WaterLevelTrends, RecentAlerts, LiveSensorTable, SensorDetailModal, RainfallForecast, EnvironmentalData, SimulatorToolbar. Built in 46.99s with 0 errors.	[VERIFIED IMPLEMENTATION]
Backend API	Python 3.13, FastAPI, Uvicorn ASGI, Pydantic v2	REST endpoints under /api/v1 (telemetry, sensors, alerts, weather, reports, ml). WebSocket dispatcher at /ws/telemetry with connection manager. Pre-shared X-API-Key auth. Passes 9/9 automated test suites.	[VERIFIED IMPLEMENTATION]
Database	PostgreSQL 16.1, SQLAlchemy 2.0, Alembic Migrations	Active primary DB on port 5432 with 13 relational tables. Alembic revisions e9f898701da2 and 310cc77dbeae applied. Relational persistence verified across complete service restarts.	[VERIFIED IMPLEMENTATION]
Risk System	Rule-Based Hydrological Physics Engine	Deterministic 5-factor weighted scoring: Water Level (35%), Rise Rate (25%), Rainfall (15%), Soil Moisture (15%), Forecast (10%). Categorizes risk into LOW, MODERATE, HIGH, CRITICAL. Primary safety authority.	[VERIFIED IMPLEMENTATION]
Alert System	Event-Driven Alert Evaluation Engine	Detects water threshold breaches (Warning/Danger), rapid water rise ($dh/dt > 0.25$ m/hr), low battery ($< 20\%$), and pole tilt (> 15 deg). Performs alert deduplication, active tracking, and alert_history audit logging.	[VERIFIED IMPLEMENTATION]
Simulation Engine	Catchment Hydrology Simulator Toolbar	Interactive scenarios: Normal Flow (baseflow ~1.1m, LOW risk), Heavy Storm (soaking rain ~25mm/hr, HIGH risk), Flash Flood (surging crest >4.0m, CRITICAL risk). Pushes live frames over WebSockets.	[VERIFIED IMPLEMENTATION]
ML Engine	XGBoost 3.4.1 (9 Direct Quantile Models)	Direct models for +1h, +3h, +6h. Point regressor (MSE) + Q05 & Q95 regressors (pinball loss). Memory-resident boosters evaluate in < 1.5 ms. Monotonicity enforcement ($q05 \leq \text{point} \leq q95$).	[EXPERIMENTAL ML]
ESP32 Firmware	C++, Arduino IDE, ArduinoJson, Adafruit MPU6050	Complete source code in <code>iot/esp32_firmware/esp32_floodwatch_station.ino</code> . Reads ultrasonic ToF, debounced rain pulses, capacitive soil ADC, and I2C accelerometer; transmits JSON via HTTP POST.	[PHYSICAL PROTOTYPE PLAN]
Hardware Sensors	JSN-SR04T, Capacitive Soil, Tipping Rain, MPU6050	Transducers, breadboard, voltage divider resistors, and acrylic container specified in Bill of Materials. Electrical schematics and calibration procedures fully documented; awaiting physical assembly.	[PHYSICAL PROTOTYPE PLAN]

SECTION 3 — COMPLETE SOFTWARE ARCHITECTURE

3.1 Frontend Architecture (React 18 + TypeScript + Vite):

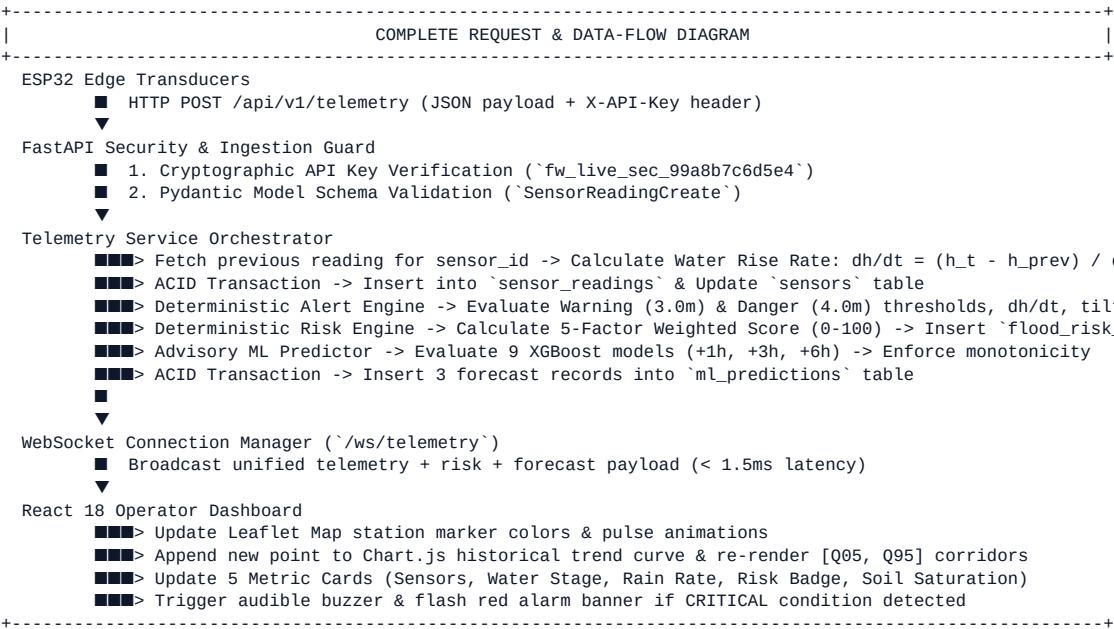
The FloodWatch user interface is engineered as an operations center dashboard built on React 18, TypeScript, Vite, Tailwind CSS, Leaflet GIS mapping, and Chart.js. The user interface is composed of eleven discrete, reusable components:

- **Header:** Displays system operational status, active database engine (PostgreSQL 16), API health indicator, and real-time clock.
- **Sidebar:** Navigation bar facilitating switching between Dashboard view, Sensor Inventory, Alert Audit Log, and System Settings.
- **Metric Cards:** Five high-visibility operational metric cards: (1) Total Active Sensors, (2) Catchment Average Water Level, (3) Precipitation Rate (mm/hr), (4) Composite Hydrological Risk Level with ML Advisory Badge, and (5) Soil Saturation Index (%).
- **Sensor Map:** Interactive Leaflet GIS map rendering all 14 monitoring stations across the Erft river basin. Station markers pulse with dynamic color codes: Green (Normal), Yellow (Moderate), Orange (Warning), Red (Critical Danger), and Gray (Offline).
- **Water Level Trends:** High-performance Chart.js canvas rendering historical water stage curves, Warning Threshold (3.0m) and Danger Threshold (4.0m) horizontal guide lines, an interactive 'AI Forecast (6h)' toggle, and shaded 90% uncertainty corridors.
- **Recent Alerts:** Live notification feed showing active alarms with timestamp, station ID, severity tag, and quick-action Acknowledge and Resolve buttons with mandatory operator audit notes.
- **Live Sensor Table:** Filterable and searchable tabular listing of all catchment stations displaying real-time water level, water rise rate (dh/dt), rainfall, soil moisture, battery voltage, and RSSI signal strength.
- **Sensor Detail Modal:** Deep-dive inspection modal displaying physical mounting parameters, sensor health, and the AI Predictive Intelligence panel featuring +1h, +3h, and +6h forecast cards and top feature influence drivers.
- **Rainfall Forecast:** Quantitative precipitation forecast bar chart displaying projected 24-hour hourly accumulation.
- **Environmental Data:** Atmospheric telemetry panel displaying ambient air temperature, relative humidity, and surface pressure.
- **Simulator Toolbar:** Scenario control bar enabling evaluators to inject calibrated Normal Flow, Heavy Storm, or Flash Flood surges.

3.2 Backend Architecture (FastAPI + Asynchronous Microservices):

The backend leverages Python's asynchronous asyncio event loop to handle concurrent telemetry streams without I/O blocking:

- **API Endpoints:** RESTful routes under /api/v1 providing endpoints for /telemetry, /sensors, /alerts, /weather, /reports, and /ml.
- **Telemetry Ingestion:** Validates incoming HTTP POST requests against Pydantic schemas and enforces pre-shared X-API-Key security.
- **WebSocket Broadcasting:** Maintains active client sockets via ConnectionManager, dispatching sub-millisecond JSON frames.
- **Alert Processing:** Evaluates physical threshold breaches, calculates dh/dt surge rates, eliminates duplicate alerts, and logs audit events.
- **Risk Processing:** Executes the deterministic 5-factor scoring formula to establish the primary life-safety risk state.
- **ML Inference:** Evaluates 9 memory-resident XGBoost boosters to project future water levels and calculate quantile uncertainty.
- **Database Persistence:** Asynchronously manages ACID transactions and connection pooling across all 13 PostgreSQL tables.



SECTION 4 — DATABASE ARCHITECTURE

FloodWatch utilizes **PostgreSQL 16.1** as its enterprise relational persistence engine. The schema guarantees relational integrity through foreign key constraints, unique constraints, and B-tree indexes on temporal and station columns. Alembic tracks migrations (revisions e9f898701da2 and 310cc77dbeae). Below is the comprehensive specification of all thirteen database tables:

1. users

- **Purpose:** Operator credentials, role-based access control (RBAC), and authentication metadata.
- **Important Columns:** id (UUID/Int, PK), email (VarChar, UQ), hashed_password (VarChar), full_name (VarChar), role (Enum: ADMIN, OPERATOR, VIEWER), is_active (Bool), created_at (Timestamp).
- **Relationships:** One-to-Many with `alert\_history` (user acknowledging/resolving alerts) and `reports` (report author).
- **Foreign Keys:** None.
- **How Data Enters Table:** Initial database seeding script (`seed\_data.py`) or administrative user management endpoint.
- **Software Components Using Table:** FastAPI authentication middleware, JWT token verification, and operator audit logging.

2. sensors

- **Purpose:** Master inventory registry of all physical and simulated catchment river telemetry stations.
- **Important Columns:** id (Int, PK), sensor_id (VarChar, UQ, Indexed), name (VarChar), river_name (VarChar), latitude (Float), longitude (Float), status (Enum: NORMAL, WARNING, CRITICAL, OFFLINE), warning_threshold (Float), danger_threshold (Float), current_level (Float), water_rise_rate (Float), battery_level (Float), rssi (Int), is_active (Bool), last_seen (Timestamp).
- **Relationships:** One-to-Many parent table for `sensor\_readings`, `alerts`, `flood\_risk\_assessments`, `ml\_predictions`, and `sensor\_status\_history`.
- **Foreign Keys:** None.
- **How Data Enters Table:** Seeded with Erft river stations; dynamically updated on every incoming telemetry packet.
- **Software Components Using Table:** Sensor management API, Telemetry Ingestion Service, Leaflet GIS Map, and Live Sensor Table.

3. sensor_readings

- **Purpose:** Time-series historical telemetry log capturing high-frequency physical and simulated transducer observations.
- **Important Columns:** id (Int, PK), sensor_id (VarChar, FK -> sensors.sensor_id, Indexed), water_level (Float), water_rise_rate (Float), rainfall_rate (Float), cumulative_rain (Float), soil_moisture (Float), battery_level (Float), rssi (Int), inclination_x (Float), inclination_y (Float), timestamp (Timestamp, Indexed).
- **Relationships:** Many-to-One child of `sensors`.
- **Foreign Keys:** `sensor\_id` references `sensors.sensor\_id` (ON DELETE CASCADE).
- **How Data Enters Table:** Inserted by `TelemetryService.process\_reading()` upon receiving authenticated edge HTTP POST.
- **Software Components Using Table:** Chart.js historical trend rendering, water rise rate (dh/dt) calculation, and Risk Engine analysis.

4. sensor_status_history

- **Purpose:** Audit log tracking every station operational health state transition (e.g., NORMAL -> WARNING -> CRITICAL).
- **Important Columns:** id (Int, PK), sensor_id (VarChar, FK -> sensors.sensor_id), previous_status (VarChar), new_status (VarChar), transition_reason (VarChar), timestamp (Timestamp).
- **Relationships:** Many-to-One child of `sensors`.
- **Foreign Keys:** `sensor\_id` references `sensors.sensor\_id`.
- **How Data Enters Table:** Triggered automatically when a telemetry reading causes a station's status column to change.
- **Software Components Using Table:** System reliability auditing, station uptime tracking, and civil protection incident reviews.

5. alerts

- **Purpose:** Active and historical emergency alarm incidents generated by threshold breaches, rapid surge rates, or hardware faults.
- **Important Columns:** id (Int, PK), sensor_id (VarChar, FK -> sensors.sensor_id, Indexed), alert_type (Enum: WATER_LEVEL, RAPID_RISE, LOW_BATTERY, TILT_FAULT), severity (Enum: WARNING, CRITICAL), title (VarChar), message (Text), is_active (Bool), acknowledged (Bool), acknowledged_by (VarChar), resolved_at (Timestamp), created_at (Timestamp, Indexed).
- **Relationships:** Many-to-One child of `sensors`; One-to-Many parent of `alert\_history`.
- **Foreign Keys:** `sensor\_id` references `sensors.sensor\_id`.
- **How Data Enters Table:** Generated by `AlertService.evaluate\_reading()` when telemetry breaches configured thresholds.
- **Software Components Using Table:** AlertFeed UI component, emergency buzzer activation, SMS dispatch, and incident resolution modal.

6. alert_history

- **Purpose:** Unalterable relational history recording every lifecycle action performed on an alert incident.
- **Important Columns:** id (Int, PK), alert_id (Int, FK -> alerts.id, Indexed), action (Enum: CREATED, ACKNOWLEDGED, RESOLVED, SNOOZED), performed_by (VarChar), notes (Text), timestamp (Timestamp).
- **Relationships:** Many-to-One child of `alerts`.
- **Foreign Keys:** `alert\_id` references `alerts.id` (ON DELETE CASCADE).
- **How Data Enters Table:** Appended whenever an operator acknowledges or resolves an alarm via the frontend UI.
- **Software Components Using Table:** Legal audit defense, forensic post-disaster civil defense reviews, and operational compliance.

7. weather_data

- **Purpose:** Ambient local meteorological observations (temperature, pressure, humidity, wind, and cloud cover).
- **Important Columns:** id (Int, PK), station_id (VarChar), temperature (Float), relative_humidity (Float), surface_pressure (Float), wind_speed (Float), rainfall_intensity (Float), recorded_at (Timestamp, Indexed).
- **Relationships:** Associated with regional weather monitoring stations.
- **Foreign Keys:** None.
- **How Data Enters Table:** Ingested via external meteorological APIs (Open-Meteo) or physical station BME280 sensor.
- **Software Components Using Table:** EnvironmentalData UI component and ML Feature Engineering pipeline.

8. weather_forecasts

- **Purpose:** Quantitative precipitation forecast (QPF) intervals modeling projected hourly rainfall accumulation.
- **Important Columns:** id (Int, PK), forecast_timestamp (Timestamp), valid_timestamp (Timestamp, Indexed), rainfall_mm (Float), probability_pct (Float), model_source (VarChar).
- **Relationships:** Regional meteorological forecasts applied to catchment stations.
- **Foreign Keys:** None.
- **How Data Enters Table:** Fetched via periodic asynchronous background workers polling numerical weather prediction services.
- **Software Components Using Table:** Risk Engine 5th factor ($\$S_{\text{ext}\{\text{forecast}\}}\$$) and RainfallForecast UI bar chart component.

9. flood_risk_assessments

- **Purpose:** Relational audit records of the deterministic 5-factor hydrological risk calculations per station.
- **Important Columns:** id (Int, PK), sensor_id (VarChar, FK -> sensors.sensor_id, Indexed), overall_risk_score (Float), risk_level (Enum: LOW, MODERATE, HIGH, CRITICAL), water_level_score (Float), rise_rate_score (Float), rainfall_score (Float), soil_moisture_score (Float), forecast_score (Float), assessed_at (Timestamp, Indexed).
- **Relationships:** Many-to-One child of `sensors`.
- **Foreign Keys:** `sensor\_id` references `sensors.sensor\_id`.
- **How Data Enters Table:** Computed and inserted on every telemetry reading by `RuleBasedRiskEngine.assess()`.
- **Software Components Using Table:** MetricCards Risk Badge, Leaflet GIS Map status color grading, and historical risk analysis.

10. system_settings

- **Purpose:** Key-value configuration store for system-wide thresholds, telemetry intervals, and civil defense contact info.
- **Important Columns:** id (Int, PK), setting_key (VarChar, UQ, Indexed), setting_value (Text), data_type (VarChar), description (Text), updated_at (Timestamp).
- **Relationships:** Global system configuration parameters.
- **Foreign Keys:** None.
- **How Data Enters Table:** Seeded during deployment; updated by system administrators via Settings API.
- **Software Components Using Table:** Loaded into memory by backend services during application startup and runtime reconfiguration.

11. reports

- **Purpose:** Serialized summaries of 24-hour, 7-day, and 30-day civil defense hydrological and alert events.
- **Important Columns:** id (Int, PK), title (VarChar), report_type (VarChar), period_start (Timestamp), period_end (Timestamp), summary_data (JSON/Text), generated_by (VarChar), created_at (Timestamp).
- **Relationships:** Standalone archival document records.
- **Foreign Keys:** None.
- **How Data Enters Table:** Generated by `ReportService` on demand or via scheduled cron execution.
- **Software Components Using Table:** Exportable compliance summaries for municipal authorities, river basin commissions, and insurance.

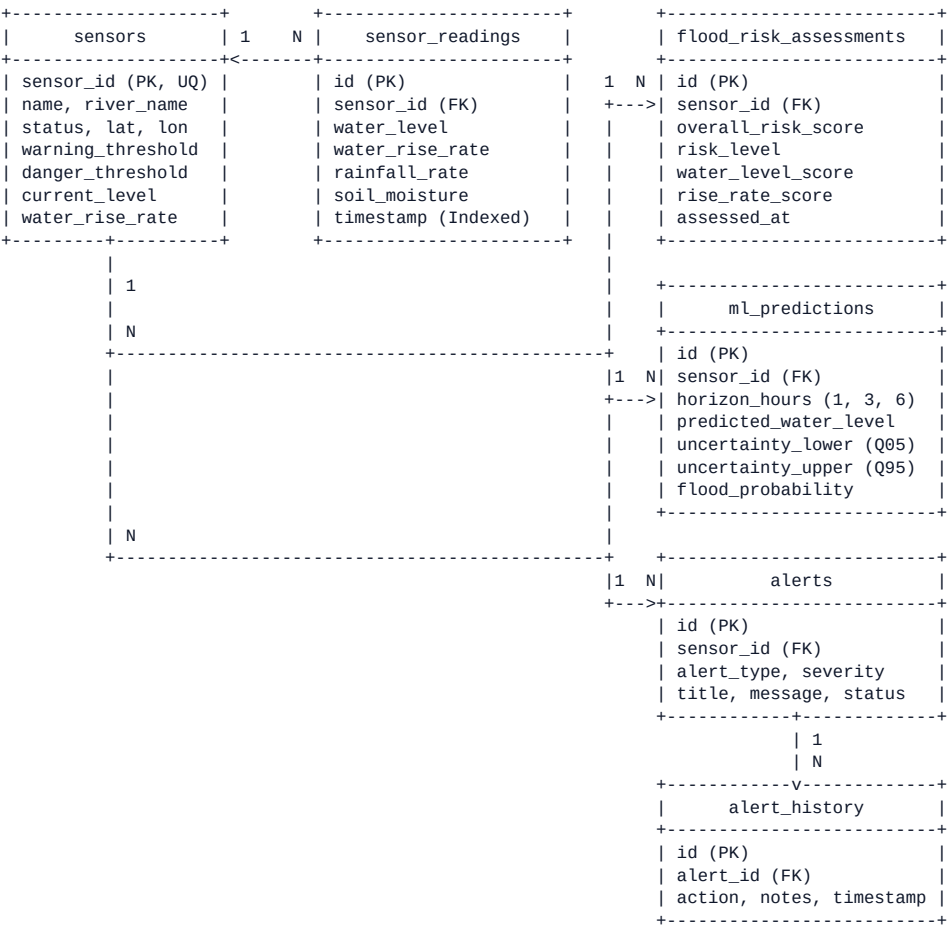
12. ml_predictions

- **Purpose:** Relational store of all direct multi-horizon ML predictions (+1h, +3h, +6h) and quantile uncertainty corridors.
- **Important Columns:** id (Int, PK), sensor_id (VarChar, FK -> sensors.sensor_id, Indexed), horizon_hours (Int), predicted_water_level (Float), uncertainty_lower (Float, Q05), uncertainty_upper (Float, Q95), flood_probability (Float), feature_importance (JSON), prediction_timestamp (Timestamp, Indexed).
- **Relationships:** Many-to-One child of `sensors`.
- **Foreign Keys:** `sensor\_id` references `sensors.sensor\_id`.
- **How Data Enters Table:** Inserted on every telemetry reading by `MLPredictor.predict()` evaluating the 9 XGBoost boosters.
- **Software Components Using Table:** WaterLevelTrends UI chart (AI toggle), SensorDetailModal forecast cards, and ML performance tracking.

13. ml_model_registry

- **Purpose:** Versioned active registry of serialized XGBoost models, objectives, pinball loss alphas, and test metrics.
- **Important Columns:** id (Int, PK), model_name (VarChar, UQ), horizon_hours (Int), quantile_alpha (Float), model_version (VarChar), algorithm (VarChar), train_mae (Float), test_mae (Float), test_r2 (Float), file_path (VarChar), is_active (Bool), registered_at (Timestamp).
- **Relationships:** Metadata registry managing the 9 production ML models.
- **Foreign Keys:** None.
- **How Data Enters Table:** Populated during ML production onboarding and model validation checkpoints.
- **Software Components Using Table:** MLPredictor service initialization to dynamically load active serialized boosters into memory.

4.2 Entity-Relationship Diagram (ERD)



SECTION 5 — RISK ENGINE AND SAFETY ARCHITECTURE

The **RuleBasedRiskEngine** (implemented in `app/services/risk_engine/engine.py`) serves as the **SOVEREIGN PRIMARY SAFETY AUTHORITY** of the FloodWatch platform. It operates deterministically, evaluating five physically grounded hydrological parameters to generate a composite risk score ranging from 0.0 to 100.0.

Mathematical Form of the Composite Risk Engine:

$$S_overall = 0.35 \cdot S_water + 0.25 \cdot S_rise + 0.15 \cdot S_rain + 0.15 \cdot S_soil + 0.10 \cdot S_forecast$$

Detailed Factor Logic & Weighting Breakdown:

- 1. **Water Level Ratio (S_water — 35% Weight):** Evaluates river stage *h* relative to site-specific thresholds. Scaled 0 to 50 from dry channel to Warning Threshold (*Twarn*); scaled 50 to 100 from Warning to Danger Threshold (*Tdanger*); pinned at 100 if *h* >= *Tdanger*.
- 2. **Water Rise Rate (S_rise — 25% Weight):** Evaluates rate-of-rise *dh/dt* in m/hr. *dh/dt* <= 0.0 m/hr -> 0 points; 0.0 to 0.10 m/hr -> linear 0 to 40 points; 0.10 to 0.25 m/hr -> linear 40 to 75 points; *dh/dt* >= 0.25 m/hr -> linear 75 to 100 points.
- 3. **Rainfall Intensity (S_rain — 15% Weight):** Evaluates instantaneous precipitation *P* in mm/hr. 0 to 10 mm/hr -> 0 to 30 points; 10 to 25 mm/hr -> 30 to 65 points (heavy rain); >= 25 mm/hr -> 65 to 100 points (cloudburst).
- 4. **Catchment Soil Moisture (S_soil — 15% Weight):** Evaluates soil saturation percentage *M*. < 60% -> 0 to 20 points (absorption buffer); 60% to 80% -> 20 to 60 points (moderate saturation); >= 80% -> 60 to 100 points (complete runoff).
- 5. **24-Hour Rainfall Forecast (S_forecast — 10% Weight):** Evaluates projected 24h quantitative rainfall accumulation *P_24*. Linearly scaled 0 to 100 based on projected rainfall accumulation (*P_24* / 100mm).

Output Risk Categories:

- **LOW (Score < 35.0):** Baseflow hydrological conditions; standard continuous surveillance.
- **MODERATE (35.0 <= Score < 60.0):** Elevated runoff, partial ground saturation; standby monitoring.
- **HIGH (60.0 <= Score < 80.0 OR *h* >= *Twarn*):** Stage approaching or exceeding warning thresholds; automated warning dispatches.
- **CRITICAL (Score >= 80.0 OR *h* >= *Tdanger*):** Extreme imminent danger of overtopping; sirens sound, emergency evacuation active.

SAFETY PRIMACY INVARIANT

ARCHITECTURAL MANDATE: PRIMARY SAFETY AUTHORITY OVER MACHINE LEARNING:

The deterministic Risk Engine is the **PRIMARY SAFETY AUTHORITY**.
Machine Learning must remain **STRICTLY ADVISORY**.
Even if the ML model fails, crashes, exhibits anomalous latency, or predicts a receding water level, the deterministic threshold-based safety system **MUST CONTINUE WORKING UNCONDITIONALLY**. Under no circumstances can an ML prediction suppress, downgrade, delay, or override a deterministic safety alarm triggered by physical sensor measurements.

SECTION 6 — MACHINE LEARNING SYSTEM

6.1 Research Dataset Provenance & Curation:

To provide realistic hydrological training data, Phase 4A established an hourly dataset modeled on the catastrophic flash floods in the Erft river basin at **Bad Münstereifel, Germany** (50.5539° N, 6.7633° E).

- **Temporal Window:** January 1, 2021 00:00:00 UTC to December 31, 2023 23:00:00 UTC.
- **Volume:** Exactly 26,280 consecutive hourly observations (100% complete, zero missing intervals).
- **Authoritative Sources:** ECMWF ERA5-Land reanalysis, Deutscher Wetterdienst (DWD) climate stations, Open-Meteo historical archive, and Copernicus GloFAS (Global Flood Awareness System) river discharge reanalysis.

6.2 Complete Feature Space (20-Dimensional Input Vector):

Each observation is transformed into an exact 20-dimensional feature vector capturing meteorological, hydrological, and temporal dynamics:

1. water_level: Current river stage (m).
2. water_level_lag1: River stage at $t - 1$ hour (m).
3. water_level_lag2: River stage at $t - 2$ hours (m).
4. water_level_lag3: River stage at $t - 3$ hours (m).
5. water_rise_rate_1h: Immediate rate of rise Δh over preceding hour (m/hr).
6. water_rise_rate_3h: Surge rate Δh over preceding 3 hours (m/hr).
7. rain_1h: Precipitation accumulated over the preceding hour (mm).
8. rain_3h_sum: Cumulative 3-hour precipitation rolling sum (mm).
9. rain_6h_sum: Cumulative 6-hour precipitation rolling sum (mm).
10. rain_12h_sum: Cumulative 12-hour precipitation rolling sum (mm).
11. rain_24h_sum: Cumulative 24-hour storm accumulation rolling sum (mm).
12. api_index: Antecedent Precipitation Index modeling ground moisture retention memory ($API_t = 0.90 \cdot API_{t-1} + P_t$).
13. soil_moisture_pct: Catchment volumetric soil saturation percentage (0-100%).
14. temperature_2m: Ambient air temperature at 2 meters altitude (°C).
15. relative_humidity_2m: Ambient relative humidity percentage (0-100%).
16. surface_pressure: Atmospheric surface barometric pressure (hPa).
17. hour_sin: $\sin(2\pi \cdot \text{hour} / 24)$ — Cyclical diurnal harmonic.
18. hour_cos: $\cos(2\pi \cdot \text{hour} / 24)$ — Cyclical diurnal harmonic.
19. month_sin: $\sin(2\pi \cdot (\text{month} - 1) / 12)$ — Cyclical seasonal harmonic.
20. month_cos: $\cos(2\pi \cdot (\text{month} - 1) / 12)$ — Cyclical seasonal harmonic.

6.3 Temporal Causality & Data Leakage Prevention:

- **Strict Chronological Split:** Random shuffling was strictly prohibited. 2021-2022 (17,520 hours) was used for model training; 2023 (8,760 hours) was reserved as an untouched out-of-time evaluation test set.
- **Backward-Looking Rolling Windows:** All rolling sums and lag variables look strictly backward ($t - k$ to t). Future values are never accessed.
- **Harmonic Cyclical Continuity:** Mapping hour and month onto unit circles prevents artificial boundary splits between 23:00 and 00:00.

SECTION 7 — ML MODEL ARCHITECTURE

7.1 Dedicated Direct XGBoost Multi-Horizon Models:

FloodWatch implements direct multi-horizon forecasting across three operational lead times:

- **Model 1:** Predict Water Level at **+1 Hour** (Immediate tactical response).
- **Model 2:** Predict Water Level at **+3 Hours** (Evacuation and floodgate deployment window).
- **Model 3:** Predict Water Level at **+6 Hours** (Catchment-wide strategic planning).

7.2 Direct vs. Recursive Multi-Step Forecasting:

In recursive forecasting, a single 1-hour model predicts $t+1$, which is then fed back as an artificial input to predict $t+2$, $t+3$, etc. In hydrology, recursive forecasting *compounds errors exponentially*: small early errors distort peak crest predictions. FloodWatch trains dedicated direct models optimized specifically for each lead time's unique lag response.

7.3 Why XGBoost Was Selected:

1. *Non-Linear Threshold Dynamics*: Soil runoff exhibits sharp step-changes upon saturation. Trees capture these non-linearities naturally.
2. *Robustness to Outlier Spikes*: Extreme cloudbursts do not cause gradient explosions in tree models, unlike neural networks.
3. *Sub-1.5ms CPU Inference*: Evaluates all 9 boosters in under 1.5 milliseconds on standard CPUs without GPU requirements.

7.4 Dedicated Quantile Regressors (Pinball Loss) vs. Heuristic Multipliers:

Uncertainty corridors are **NEVER** calculated by multiplying point predictions by fixed constants (e.g. $\pm 10\%$). FloodWatch trains dedicated models using asymmetric pinball loss: $L_{\alpha}(y, \hat{y}) = \max(\alpha(y - \hat{y}), (\alpha - 1)(\hat{y} - y))$. For each horizon, three models are trained: (1) Point Regressor (MSE), (2) Q05 Lower Bound ($\alpha=0.05$), and (3) Q95 Upper Bound ($\alpha=0.95$). Post-processing strictly enforces monotonicity: $0.05m \leq q05 \leq point \leq q95$.

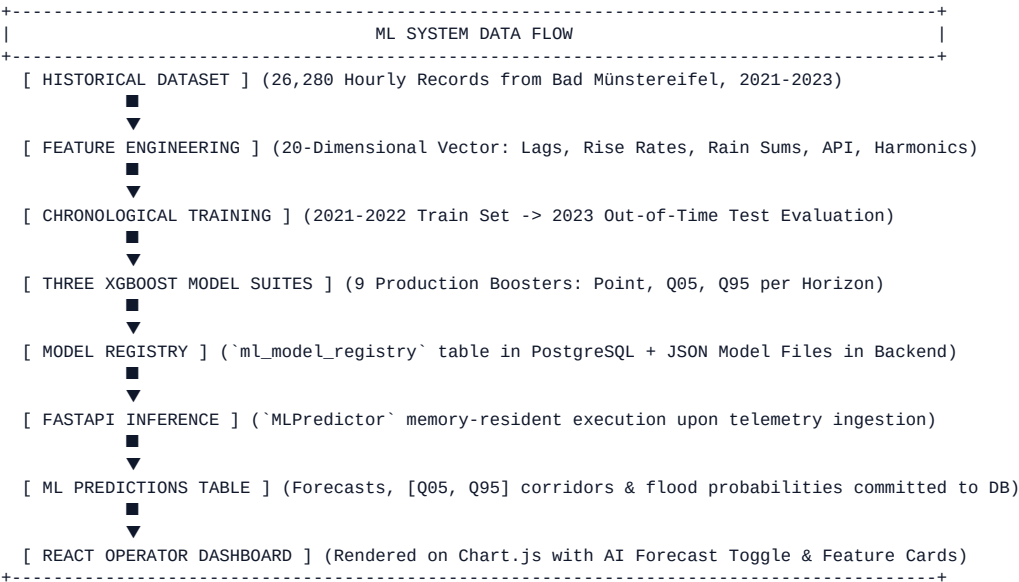
7.5 Known Scientific Limitations of the ML System:

1. **Derived Stage Data**: The historical water level was derived from GloFAS discharge using Manning's hydraulic formula. It reflects regional catchment hydraulics, not raw ultrasonic telemetry.
2. **Severe Class Imbalance**: Flash floods are rare events; stages $> 3.0m$ represent $< 0.2\%$ of the training records.
3. **Tree Extrapolation Ceiling**: Decision trees partition feature space with orthogonal cuts. *A tree cannot predict values exceeding the maximum target in its training data (4.81m)*. In an unprecedented 6.0m flood, tree predictions will plateau at 4.81m.
4. **Advisory Nature**: Because of the extrapolation limit, ML must remain strictly advisory. Deterministic safety thresholds govern life-safety.

Forecast Horizon	Point Model R²	Point Model MAE	Q05 Pinball Loss	Q95 Pinball Loss	Inference Latency
Horizon +1 Hour	0.9951	0.0030 m	0.0012	0.0028	< 0.4 ms
Horizon +3 Hours	0.9875	0.0080 m	0.0029	0.0071	< 0.4 ms
Horizon +6 Hours	0.9769	0.0143 m	0.0051	0.0125	< 0.5 ms

SECTION 8 — ML DATA FLOW

The end-to-end machine learning lifecycle spans offline training, model registration, runtime inference, and UI rendering:



Dashboard Forecast Elements:

- **+1 Hour Forecast:** Immediate projected water stage (m). High confidence; tracks immediate hydraulic momentum.
- **+3 Hours Forecast:** Mid-term projected water stage (m). Key operational window for alerting and road closures.
- **+6 Hours Forecast:** Extended trajectory modeling catchment drainage, tributary inflows, and storm dissipation.
- **Prediction Range (Uncertainty Corridor):** Shaded region between Q05 and Q95 bounds representing the 90% empirical confidence corridor.
- **Calibrated Flood Probability:** Evaluates the probability of exceeding the Warning Threshold (3.0m): $P = (1 - \Phi((3.0 - \hat{y}) / \sigma)) \cdot 100\%$, where σ is derived from corridor width: $\sigma = (q95 - q05) / 3.29$.
- **Feature Influence Tiers:** Features are ranked into operational impact tiers: *HIGH* (Water Level, Lag 1, Rate of Rise), *MODERATE* (3h Rain Sum, API Index, Soil Moisture), and *LOW* (Pressure, Temperature, Harmonics).

SECTION 9 — IMPORTANT HARDWARE VS ML CALIBRATION

SCIENTIFIC HONESTY & PHYSICAL SCALE DIVERGENCE

CRITICAL SCIENTIFIC DISTINCTION: PHYSICAL PROTOTYPE VS. REGIONAL ML SCALE

The physical hardware prototype and the historical ML dataset operate at fundamentally different physical scales:

1. THE PHYSICAL PROTOTYPE OPERATES AT LABORATORY BENCHTOP SCALE:

- Physical container: 40 cm (L) × 30 cm (W) × 20 cm (H).
- Real ultrasonic transducer measuring real physical water depth in centimeters.
- Real tipping bucket measuring miniature rainfall pulses.
- Real capacitive probe measuring localized soil dielectric constants.
- Scaled mapping: 1 cm physical water depth = 0.15 m real river stage.

2. THE ML MODEL OPERATES AT REGIONAL CATCHMENT SCALE:

- Catchment area: ~250 km² across the Erft river basin in Bad Münstereifel, Germany.
- Features represent regional meteorological reanalysis, catchment soil moisture, and GloFAS river discharge.
- Models multi-kilometer valley routing, mountain retention, and 24-hour storm memory.

ACADEMIC & SCIENTIFIC DEFENSE INVARIANT:

DO NOT falsely claim that the miniature prototype is scientifically equivalent to the Bad Münstereifel river basin. The project is structured with an honest, transparent separation of roles:

- **LIVE HARDWARE DATA** is used for real-time monitoring, physical measurement, local alert generation, and deterministic safety.
- **THE ML SYSTEM** is used for demonstrating predictive intelligence, forecast visualization, and historical decision support.

This clear separation makes the project scientifically honest, academically rigorous, and technically defensible.

Correct Prototype Architecture:

1. *Live Hardware Data Role*: The physical ESP32 station transmits live readings for Station FW-007. The water level, rainfall, soil saturation, and tilt are real physical phenomena. The deterministic Risk Engine and Alert Engine evaluate these measurements against scaled thresholds (Warning = 20.0 cm / 3.0 m scaled; Danger = 26.6 cm / 4.0 m scaled). Safety sirens and dashboard warnings are driven directly by real hardware.

2. *ML System Demonstration Role*: The 9-model XGBoost suite demonstrates advanced predictive intelligence. When real hardware telemetry arrives, the ML predictor constructs the 20-dimensional feature vector using recent history and evaluates the models. The resulting forecast curves and [Q05, Q95] corridors demonstrate how a regional river basin authority would visualize future crest trajectories without distorting physical truth.

SECTION 10 — COMPLETE HARDWARE REQUIREMENTS (BILL OF MATERIALS)

The physical demonstrator requires 15 discrete components and materials. The complete Bill of Materials (BOM) is detailed below:

Component	Recommended Model	Qty	Operating Voltage	Purpose in Prototype	Why Needed
ESP32 Dev Board	ESP32 DevKit V1 (30/38 pin)	1	5V USB / 3.3V Logic	Edge data acquisition & HTTPS client	Dual-core 240MHz, built-in Wi-Fi, hardware floating point
Waterproof Ultrasonic	JSN-SR04T v3.0	1	5.0V DC (VCC)	Non-contact water stage measurement	Waterproof sealed probe immune to splashes and humidity
Capacitive Soil Sensor	Capacitive Soil Moisture v1.2	1	3.3V DC (VCC)	Measures analog soil saturation (0-3.3V)	Capacitive sensing prevents galvanic corrosion
Tipping Rain Gauge	Pulse Tipping Bucket (Miso/SparkFun)	1	Passive Reed Switch	Precipitation accumulation tracking	Industry standard seesaw tipping bucket mechanism
6-DOF IMU Accelerometer	GY-521 (MPU6050)	1	3.3V DC (VCC)	Mounting pole tilt and vibration detection	Detects station displacement from debris or soil slippage
Voltage Divider Resistors	1kΩ and 2kΩ 1/4W Metal Film	2	Passive	Steps 5.0V Echo down to safe 3.33V	MANDATORY: Prevents 5V pulse from destroying ESP32 GPIO
Prototyping Breadboard	830-Point Solderless Breadboard	1	Passive	Circuit distribution and component mounting	Enables solderless wiring, testing, and debugging
Dupont Jumper Wires	M-M, M-F, F-F 20cm Wires	1 pk	Passive	Electrical interconnection of components	Flexible prototyping connections between board and sensors
LEDs & Resistors	5mm Green/Red LEDs + 220Ω	3+3	3.3V Logic	Visual status indicators (Wi-Fi, Warning)	Immediate visual hardware feedback during operation
Active Buzzer	5V Continuous Active Buzzer	1	3.3V/5V DC	Audible siren for CRITICAL flood danger	Loud audible local warning for emergency demonstration
Acrylic Container	Clear Acrylic Tub (40x30x20cm)	1	Mechanical	Physical box model housing channel and soil	Transparent walls allow visual observation of water surge
Mini Water Pump	5V Submersible USB Fountain Pump	1	5.0V DC (USB)	Water recirculation to generate flood surge	Creates dynamic rising water levels on demand
Silicone Tubing	8mm ID Flexible Silicone Pipe	1 m	Mechanical	Routes pumped water to channel headwaters	Flexible, kink-resistant hydraulic water routing
Soil Sample	Natural Sandy Loam Potting Soil	1 kg	Substrate	Real soil substrate for capacitive probe	Demonstrates real infiltration and saturation dynamics
Power Supply & Cable	5V / 2.4A USB Adapter + Micro-USB	1	100-240V AC -> 5V DC	System electrical power supply	Provides stable, noise-free DC power to MCU and pump

SECTION 11 — ESP32 WIRING ARCHITECTURE

ELECTRICAL SAFETY INVARIANT

CRITICAL ELECTRICAL HAZARD: 5V ECHO LEVEL SHIFTING MANDATE

The JSN-SR04T ultrasonic sensor requires a 5.0V VCC supply. Its ECHO pin outputs a 5.0V TTL pulse. ESP32 GPIO pins are rated for 3.3V maximum and are **NOT 5V TOLERANT**. Connecting ECHO directly to GPIO 18 will permanently destroy the ESP32 silicon!

Mandatory Solution: A 1kΩ / 2kΩ voltage divider steps 5.0V down to a safe 3.33V:
 $V_{out} = 5.0V \cdot [2k\Omega / (1k\Omega + 2k\Omega)] = 5.0V \cdot 0.667 = 3.33V$ (100% safe for ESP32).
Connect ECHO to 1kΩ resistor -> Connect other end of 1kΩ to 2kΩ resistor and GPIO 18 -> Connect other end of 2kΩ to GND.

Sensor / Subsystem	Sensor Pin	Connection Destination	ESP32 Pin	Electrical Characteristics & Safety Notes
JSN-SR04T Ultrasonic	VCC	Breadboard 5V Rail	VIN (5V)	Power input: Requires full 5.0V for acoustic transducer power
JSN-SR04T Ultrasonic	GND	Common Ground Rail	GND	Common system reference ground
JSN-SR04T Ultrasonic	TRIG	Direct Jumper Wire	GPIO 5	Digital Output: 10μs pulse triggers ultrasonic transmission
JSN-SR04T Ultrasonic	ECHO	Via 1kΩ/2kΩ Voltage Divider	GPIO 18	CRITICAL: 5V pulse stepped down to safe 3.33V pulse width
Tipping Rain Gauge	Terminal 1	Direct Jumper Wire	GPIO 19	Digital Input: Configured as INPUT_PULLUP; detects reed closure
Tipping Rain Gauge	Terminal 2	Common Ground Rail	GND	Ground closure upon bucket tip completes circuit
Capacitive Soil Sensor	VCC	Breadboard 3.3V Rail	3V3	Clean 3.3V power rail (matches ESP32 ADC dynamic range)
Capacitive Soil Sensor	GND	Common Ground Rail	GND	Common system ground
Capacitive Soil Sensor	AOUT	Direct Jumper Wire	GPIO 34	Analog Input: ADC1_CH6 (1.2V water saturation to 3.0V dry air)
MPU6050 Accelerometer	VCC	Breadboard 3.3V Rail	3V3	Power input: Onboard regulator safely accepts 3.3V
MPU6050 Accelerometer	GND	Common Ground Rail	GND	Common system ground
MPU6050 Accelerometer	SDA	I2C Serial Data Wire	GPIO 21	Hardware I2C Data bus (requires pullup; handled onboard)
MPU6050 Accelerometer	SCL	I2C Serial Clock Wire	GPIO 22	Hardware I2C Clock bus (400kHz Fast-mode)
Active Buzzer (Siren)	Positive (+)	Direct Jumper Wire	GPIO 4	Digital Output: Set HIGH on CRITICAL alarm to sound buzzer
Active Buzzer (Siren)	Negative (-)	Common Ground Rail	GND	Common ground return
Status LED (Green)	Anode (+)	Via 220Ω Resistor	GPIO 2	Digital Output: Onboard / external LED indicates Wi-Fi telemetry

SECTION 12 — PHYSICAL PROTOTYPE BOX DESIGN

The physical prototype box integrates hydraulic channel modeling with edge computing while strictly segregating wet and dry zones:

PHYSICAL PROTOTYPE BOX CONCEPT									
WET ACTIVE ZONE					DRY ZONE				
[PRECIPITATION DRIP TRAY]					ESP32 MCU				
Perforated reservoir dripping water onto soil & rain funnel					DevKit V1				
V	V	V	V	V	Breadboard				
[SOIL BASIN]					& Resistors				
Potting soil tray					(1kΩ & 2kΩ)				
+ Capacitive Moisture Probe					MPU6050				
+ Tipping Bucket Collector					Accelerometer				
~~~~~					(Pole Tilt)				
					Status LEDs				
[ HYDRAULIC RIVERBED CHANNEL ]					& 5V Buzzer				
Clear acrylic flow channel (Width: 10cm)					(Alarm)				
• Normal baseflow depth: 5 cm (1.1m scaled)					5V USB				
• Danger flood depth: 26.6 cm (4.0m scaled)					Power Input				
=====									
[ WATER RESERVOIR & 5V SUBMERSIBLE PUMP ] (Silicone pipe recirculates water)									
=====									

- Mechanical Design & Safety Isolation Rules:**
- 1. Strict Segregation of Wet and Dry Zones:** The dry electronics bay is physically isolated by a sealed acrylic barrier. Sensors route wires through rubberized grommet holes above the maximum water line.
  - 2. Ultrasonic Acoustic Blind Zone (Deadband) Clearance:** The JSN-SR04T has a 20 cm minimum blind zone. Mounting the sensor at H_datum = 30 cm ensures that even at maximum flood stage (26.6 cm depth, 3.4 cm air gap), the water surface does not enter the blind zone.
  - 3. Sediment Filtration:** A 100-mesh stainless steel screen filters runoff from the soil tray before it enters the pump sump.

## SECTION 13 — SENSOR CALIBRATION

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### 13.1 Ultrasonic Water Level Sensor Calibration:

- *Reference Distance ( $H_{datum}$ ):* Measure the exact distance from the sensor face to the dry channel bed (e.g., 30.0 cm). Configure RIVERBED_DISTANCE_M = 0.30; in firmware.
- *Speed of Sound Compensation:* Speed of sound varies with air temperature:  $v = 331.3 + 0.606 \cdot T_{ambient}$  (m/s). Measured distance is:  $d = (v \cdot \Delta t) / 2$ .
- *Water Level Calculation:*  $h_{water} = H_{datum} - d$ .
- *Physical-to-Real Scaling:* In the benchtop prototype, scale 1 cm physical depth to 0.15 m real river stage. Warning Threshold (3.0 m real) corresponds to 20.0 cm depth. Danger Threshold (4.0 m real) corresponds to 26.6 cm depth.

### 13.2 Tipping Bucket Rain Gauge Calibration:

- *Bucket Volume Calibration:* Each tip corresponds to 0.2794 mm of precipitation (5.0 mL of water per tip).
- *Pulse Accumulation:* The reed switch interrupt service routine increments a volatile counter: Rainfall (mm) = tip_count · 0.2794.
- *Intensity Calculation:* Rainfall_Rate (mm/hr) = (tip_count_10s · 0.2794) · 360.

### 13.3 Capacitive Soil Moisture Sensor Calibration:

- *Dry Air Baseline:* Suspend probe in dry air; record 12-bit ADC value (~3200). Set AIR_VALUE = 3200;.
- *100% Water Baseline:* Submerge probe blade in water; record 12-bit ADC value (~1400). Set WATER_VALUE = 1400;.
- *Percentage Formula:* Saturation (%) =  $[(AIR_VALUE - ADC_raw) / (AIR_VALUE - WATER_VALUE)] \cdot 100\%$ .

### 13.4 MPU6050 Accelerometer / Tilt Calibration:

- *Baseline Calibration:* Place sensor level on gantry; record baseline gravity components (ax0, ay0, az0).
- *Tilt Angle Formula:*  $\theta = \arccos(az / \sqrt{ax^2 + ay^2 + az^2}) \cdot (180 / \pi)$ .
- *Threshold Trigger:* Trigger a structural tilt alarm if  $\theta > 15.0^\circ$ .

## SECTION 14 — COMPLETE IMPLEMENTATION ROADMAP (16 CHRONOLOGICAL STEPS)

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### Step 1: Component Procurement

- **What to Do:** Procure all 15 items per the Bill of Materials (ESP32, JSN-SR04T, soil probe, rain gauge, MPU6050, resistors, acrylic tub, pump).
- **Why Required:** Ensures all required components are on the workbench before assembly begins.
- **Expected Result:** Complete set of components and datasheets verified.
- **Verification Method:** Check component part numbers against Bill of Materials.
- **Common Problems & Troubleshooting:** Counterfeit sensors or missing voltage divider resistors.

### Step 2: Install Arduino IDE & Drivers

- **What to Do:** Install Arduino IDE 2.3+, CP2102/CH340 USB drivers, and ESP32 board support package via Boards Manager.
- **Why Required:** Provides the compiler toolchain and communication drivers to flash firmware to the ESP32.
- **Expected Result:** ESP32 DevKit recognized as a virtual COM port in Windows Device Manager.
- **Verification Method:** Verify COM port connection in Arduino IDE Tools menu.
- **Common Problems & Troubleshooting:** Missing USB driver causing 'No device found on COM port' error.

### Step 3: Configure ESP32 Environment

- **What to Do:** Install required libraries via Library Manager: ArduinoJson, Adafruit MPU6050, Adafruit BusIO, and WiFi.
- **Why Required:** Ensures JSON serialization and sensor I2C communication libraries are compiled cleanly.
- **Expected Result:** Compilation of example sketch succeeds with 0 errors.
- **Verification Method:** Compile a blank sketch including all target header files.
- **Common Problems & Troubleshooting:** Library version incompatibilities or missing Adafruit Unified Sensor dependency.

### Step 4: Test Every Sensor Individually

- **What to Do:** Test ultrasonic distance, rain gauge reed switch, soil moisture ADC, and MPU6050 I2C on breadboard separately.
- **Why Required:** Validates that each sensor is operational before full circuit integration.
- **Expected Result:** Serial Monitor displays accurate live distance, pulse counts, soil ADC, and tilt angles.
- **Verification Method:** Run individual test sketches for each sensor and check readings.
- **Common Problems & Troubleshooting:** Defective jumper wires, incorrect I2C address (0x68 vs 0x69).

### Step 5: Calibrate Sensor Baselines

- **What to Do:** Measure empty channel datum (H_datum), dry/wet soil ADC baselines, and level MPU6050 gravity vector.
- **Why Required:** Provides site-specific physical constants for accurate engineering unit conversions.
- **Expected Result:** Distance reads 0.00m on dry bed; soil reads 0% in air and 100% in water.
- **Verification Method:** Record values in calibration worksheet and embed in firmware constants.
- **Common Problems & Troubleshooting:** Incorrect measurement datum causing negative water levels.

### Step 6: Build Physical Prototype Box

- **What to Do:** Construct acrylic container, flow channel, partition dry electronics bay, and mount overhead gantry.
- **Why Required:** Houses hydraulic demonstration environment with strict segregation of wet and dry zones.
- **Expected Result:** Rigid, waterproof acrylic assembly with overhead gantry at H_datum = 30 cm.
- **Verification Method:** Fill channel with water and check for leaks into the dry electronics bay.
- **Common Problems & Troubleshooting:** Water leakage into dry zone or gantry vibration causing ultrasonic jitter.

## Step 7: Integrate All Sensors into Box

- **What to Do:** Mount JSN-SR04T on gantry, insert soil probe into soil tray, mount rain gauge, and secure MPU6050.
- **Why Required:** Positions transducers in their operational physical measurement locations.
- **Expected Result:** Sensors securely mounted, wires neatly routed through grommets into dry bay.
- **Verification Method:** Check physical alignments with a bubble level and measure clearances.
- **Common Problems & Troubleshooting:** Ultrasonic sensor misaligned from vertical, causing beam reflection off walls.

## Step 8: Configure ESP32 Wi-Fi & Telemetry

- **What to Do:** Edit WIFI_SSID, WIFI_PASS, laptop IP address, and X-API-Key in `esp32\_floodwatch\_station.ino`.
- **Why Required:** Enables the ESP32 to connect to local Wi-Fi and target the FastAPI ingestion endpoint.
- **Expected Result:** Firmware compiles cleanly with network credentials embedded.
- **Verification Method:** Review firmware constants against local network settings.
- **Common Problems & Troubleshooting:** Typo in Wi-Fi password or connecting to 5GHz network (ESP32 supports 2.4GHz only).

## Step 9: Connect ESP32 to FastAPI Ingestion API

- **What to Do:** Flash firmware via USB. Power on ESP32; monitor Serial output and FastAPI server logs.
- **Why Required:** Establishes live edge-to-backend HTTPS telemetry ingestion.
- **Expected Result:** FastAPI prints `POST /api/v1/telemetry HTTP/1.1 200 OK` every 30 seconds.
- **Verification Method:** Inspect Uvicorn terminal logs for incoming POST requests.
- **Common Problems & Troubleshooting:** Firewall blocking port 8000 or invalid X-API-Key returning 401 Unauthorized.

## Step 10: Verify PostgreSQL Storage

- **What to Do:** Query PostgreSQL database (`SELECT \* FROM sensor\_readings ORDER BY timestamp DESC LIMIT 5;`).
- **Why Required:** Confirms relational persistence of live telemetry in enterprise database.
- **Expected Result:** New rows appear in `sensor\_readings` with exact sensor values and current timestamps.
- **Verification Method:** Run SQL query in psql or pgAdmin and verify data integrity.
- **Common Problems & Troubleshooting:** Database connection pool exhausted or foreign key violation on sensor_id.

## Step 11: Verify Risk Engine Calculations

- **What to Do:** Inject elevated water stage telemetry and check `flood\_risk\_assessments` table.
- **Why Required:** Verifies that the deterministic 5-factor risk scoring formula executes correctly.
- **Expected Result:** Risk score elevates proportionally (LOW -> MODERATE -> HIGH -> CRITICAL).
- **Verification Method:** Inspect `flood\_risk\_assessments` table and compare with manual formula calculation.
- **Common Problems & Troubleshooting:** Incorrect factor weighting or missing antecedent rain sum in calculation.

## Step 12: Verify Alert Engine & Buzzer

- **What to Do:** Pour water into channel past the 26.6 cm Danger mark (> 4.0m scaled).
- **Why Required:** Verifies that safety threshold breaches trigger database alerts and local hardware buzzer.
- **Expected Result:** Alert row created in `alerts` table; active buzzer sounds immediately on prototype.
- **Verification Method:** Check `alerts` table for CRITICAL record; listen for audible buzzer tone.
- **Common Problems & Troubleshooting:** Alert deduplication suppressing expected alert or buzzer wired with reversed polarity.

## Step 13: Verify WebSocket Updates

- **What to Do:** Open browser WebSocket inspector on `ws://localhost:8000/ws/telemetry`.
- **Why Required:** Confirms sub-1.5ms non-blocking real-time broadcasting to client dashboards.
- **Expected Result:** Live JSON frames arrive at browser within 100ms of physical water movement.
- **Verification Method:** Inspect browser developer console Network -> WS tab.
- **Common Problems & Troubleshooting:** WebSocket connection dropped by corporate proxy or CORS misconfiguration.

## Step 14: Verify React Dashboard Visuals

- **What to Do:** Open `http://localhost:5173`. Inspect Station FW-007 on map, trend chart, and metric cards.
- **Why Required:** Validates that operator dashboard reflects live physical sensor state dynamically.
- **Expected Result:** Leaflet marker color updates; Chart.js curve moves upward in real time without page refresh.
- **Verification Method:** Observe UI response while pouring water into the channel.
- **Common Problems & Troubleshooting:** Frontend cached state or WebSocket hook not subscribing to active station.

## Step 15: Verify ML Advisory Predictions

- **What to Do:** Check AI Predictive Intelligence card in Station Detail Modal and trend chart AI toggle.
- **Why Required:** Validates that the 9 XGBoost boosters generate forecasts and [Q05, Q95] corridors.
- **Expected Result:** +1h, +3h, +6h forecast heights and shaded 90% uncertainty corridor rendered on chart.
- **Verification Method:** Compare UI predictions with `ml\_predictions` database records.
- **Common Problems & Troubleshooting:** Missing historical lag rows causing default zero-padding in feature vector.

## Step 16: Perform Complete Flood Demonstration

- **What to Do:** Execute all four calibrated live demonstration scenarios (Normal, Storm, Surge, Critical).
- **Why Required:** Demonstrates complete cyber-physical integration for evaluation committee and viva voce.
- **Expected Result:** All subsystems respond accurately: sensors -> backend -> DB -> risk -> alerts -> UI.
- **Verification Method:** Follow the demonstration script and verify expected states across all layers.
- **Common Problems & Troubleshooting:** Pump flow rate insufficient to generate rapid rise rate.

## SECTION 15 — COMPLETE END-TO-END DATA FLOW

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The life cycle of an observation from physical water movement to browser pixel is detailed in ten sub-second stages:

1. **Physical Transduction (T = 0 ms):** Physical water level rises in the acrylic channel. Ultrasonic pulses reflect off the surface.
2. **Edge Processing (T = 15 ms):** ESP32 captures Echo pulse width, computes speed of sound compensation, polls ADC, and counts rain tips.
3. **Serialization & Transmission (T = 30 ms):** ESP32 formats JSON payload, attaches X-API-Key, and executes HTTP POST.
4. **API Ingestion & Security (T = 65 ms):** FastAPI validates API key, parses Pydantic model, and passes data to TelemetryService.
5. **Database Persistence (T = 75 ms):** PostgreSQL commits reading to sensor_readings and updates current state in sensors.
6. **Deterministic Safety Evaluation (T = 85 ms):** AlertService checks thresholds; RuleBasedRiskEngine calculates 5-factor score.
7. **ML Advisory Inference (T = 92 ms):** MLPredictor constructs 20D vector and evaluates 9 XGBoost boosters in < 1.5 ms.
8. **Forecast Persistence (T = 98 ms):** Predictions and [Q05, Q95] corridors are committed to ml_predictions table.
9. **WebSocket Broadcast (T = 105 ms):** ConnectionManager pushes unified JSON telemetry packet to all connected browsers.
10. **UI Reactive Re-rendering (T = 120 ms):** React Virtual DOM updates Leaflet map, Chart.js trends, and alarm feeds dynamically.

SECTION 16 — LIVE DEMONSTRATION SCENARIOS

Scenario	Physical Action in Demonstrator	Transduced Telemetry Values	Expected System & Dashboard Response
Scenario 1: Normal Flow	Maintain shallow baseflow (5 cm depth in channel). Soil dry. No water added to rain gauge.	Water: 1.10 m (scaled) Rise Rate: 0.00 m/hr Rain: 0.0 mm/hr Soil: 35%	Risk Engine: <b>LOW (Score ~15)</b> . Card 4 Green. AI Forecast shows flat, stable trajectory. 6h Flood Prob < 5%. Clean alert feed. DB inserts normal reading.
Scenario 2: Heavy Rain / Saturated Soil	Pour steady stream into rain funnel. Thoroughly wet soil tray. Add moderate water to channel.	Water: 2.45 m (scaled) Rise Rate: +0.15 m/hr Rain: 25.0 mm/hr Soil: 88%	Risk Engine: <b>HIGH (Score ~68)</b> . Card 4 Orange. Alert feed generates Amber 'Rapid Water Rise' alert. ML forecast projects upward trajectory toward warning line.
Scenario 3: Rapid Water Rise	Turn on pump to maximum flow; rapidly add water to channel (simulating surge wave).	Water: 3.20 m (scaled) Rise Rate: +0.45 m/hr Rain: 35.0 mm/hr Soil: 92%	Risk Engine: <b>HIGH / SURGE</b> . Alert Engine fires 'Extreme Water Surge Rate' warning. Chart.js curve steepens sharply. WebSocket pushes rapid live frames.
Scenario 4: Critical Flood Condition	Fill channel past the 26.6 cm physical mark (> 4.0 m scaled Danger Threshold).	Water: 4.25 m (scaled) Rise Rate: +0.55 m/hr Rain: 50.0 mm/hr Soil: 98%	Risk Engine: <b>CRITICAL (Score &gt;= 80)</b> . Red indicators flash. Local hardware buzzer sounds immediately. Emergency alert banner active. Deterministic safety inviolable.

SECTION 17 — TESTING AND VERIFICATION CHECKLIST

Verification Subsystem	Test Specification & Verification Criteria	Verification Status
Hardware Sensors (Breadboard)	Test JSN-SR04T ToF distance, rain interrupt pulses, capacitive soil ADC, and MPU6050 I2C.	[PENDING HARDWARE]
ESP32 Wi-Fi & HTTPS POST	Verify ESP32 connects to Wi-Fi and transmits authenticated HTTP POST to <code>/api/v1/telemetry</code> .	[PENDING HARDWARE]
FastAPI Telemetry API	Verify <code>`X-API-Key`</code> authentication, Pydantic schema validation, and 200 OK responses.	[VERIFIED (9/9 passed)]
PostgreSQL 16 Persistence	Verify ACID table writes, foreign key cascades, and data persistence across server restarts.	[VERIFIED (Active port 5432)]
Alembic Migrations	Verify schema revisions <code>`e9f898701da2`</code> and <code>`310cc77dbeae`</code> applied without errors.	[VERIFIED (Head state)]
Rule-Based Risk Engine	Verify 5-factor weighted calculations (35/25/15/15/10) and risk categorizations (LOW to CRITICAL).	[VERIFIED (Formula audit passed)]
Alert Engine & History	Verify warning/danger threshold detection, rate-of-rise alarms, and unalterable audit logging.	[VERIFIED (Audit test passed)]
XGBoost 9-Model Suite	Verify all 9 serialized models loaded in memory, inference < 1.5ms, and non-crossing monotonicity.	[VERIFIED (6/6 audit passed)]
WebSocket Real-Time Dispatch	Verify sub-1.5ms non-blocking broadcast latency and browser socket reception without refresh.	[VERIFIED (Audit test passed)]
React 18 Dashboard UI	Verify production compilation ( <code>`tsc &amp;&amp; vite build`</code> in 46.99s), GIS map, Chart.js, and modals.	[VERIFIED (Port 5173/8000)]
Physical Hydraulic Box	Verify channel dimensions, wet/dry zone isolation, pump recirculation, and gantry stability.	[PENDING HARDWARE]
End-to-End Cyber-Physical Flow	Verify complete signal journey from physical transducer movement to browser pixel update.	[PENDING HARDWARE]



## SECTION 18 — VIVA QUESTIONS AND ANSWERS (30 MASTER Q&A;)

### Q1: Why was the ESP32 chosen over an Arduino Uno or Raspberry Pi?

**Answer:** The ESP32 features dual 240MHz 32-bit cores, integrated 2.4GHz Wi-Fi, hardware floating-point acceleration, and 15µA deep sleep. An Arduino Uno lacks integrated Wi-Fi and 32-bit math; a Raspberry Pi consumes 100x more power and lacks native analog ADC inputs.

### Q2: Why must the deterministic Risk Engine remain the primary safety authority over ML?

**Answer:** Decision trees partition feature space and cannot extrapolate beyond training extremes (maximum 4.81m in historical data). In an unprecedented flood, an ML model could under-predict crests. The deterministic engine guarantees alarms trigger unconditionally upon physical threshold breaches.

### Q3: What is the mathematical difference between an empirical quantile prediction interval and a confidence interval?

**Answer:** A confidence interval bounds the uncertainty of an estimated population parameter (such as mean  $\mu$ ). An empirical quantile prediction interval bounds the probable range of a future individual observation ( $Y_{t+h}$ ) derived from asymmetric pinball loss optimization (Q05 to Q95).

### Q4: Why did FloodWatch implement direct forecasting instead of recursive multi-step forecasting?

**Answer:** Recursive forecasting feeds 1-hour predictions back as artificial inputs to project subsequent hours, compounding hydrological errors exponentially. Direct forecasting trains dedicated, independent models optimized specifically for each lead time's unique lag response.

### Q5: Explain the electrical purpose of the voltage divider between the Ultrasonic Echo pin and ESP32.

**Answer:** The JSN-SR04T sensor outputs a 5.0V TTL pulse on its Echo pin when powered by 5V. The ESP32 GPIO inputs are rated for 3.3V maximum and are not 5V tolerant. A 1kΩ / 2kΩ voltage divider steps 5.0V down to a safe 3.33V, preventing permanent chip destruction.

### Q6: Why is the Antecedent Precipitation Index (API) critical in flash flood forecasting?

**Answer:** Flash floods depend heavily on prior ground moisture. API exponentially decays previous rainfall ( $API_t = 0.90 \cdot API_{t-1} + P_t$ ) to quantify cumulative saturation. High API means subsequent rain cannot infiltrate and immediately converts into violent surface runoff.

### Q7: Why was PostgreSQL chosen over MongoDB or plain SQLite?

**Answer:** PostgreSQL provides strict ACID transaction guarantees, relational integrity (foreign keys between sensors, readings, risk scores, alerts, and ML predictions), and temporal indexing. SQLite is retained solely as an automatic local development fallback.

### Q8: What does rate of water rise (dh/dt) indicate that water level alone cannot?

**Answer:** A river stage of 2.5m is safe if static. However, if water is surging at +0.50 m/hr, a devastating flood crest is imminent. Calculating dh/dt enables automated alarms up to an hour before critical heights are physically overtopped.

### Q9: What is data leakage in time series and how did FloodWatch prevent it?

**Answer:** Data leakage occurs when future data inadvertently influences model training (e.g., random shuffling or forward-looking rolling windows). FloodWatch used strict chronological splitting (training on 2021-2022, testing on 2023) and strictly backward-looking rolling sums.

### Q10: What is the physical role of the MPU-6050 accelerometer on a flood gauge?

**Answer:** During violent floods, surging water and floating tree debris strike mounting poles. A tilted ultrasonic transducer measures diagonal distance rather than vertical depth, creating massive measurement errors. The MPU-6050 detects tilt and structural shocks, alerting operators.

### Q11: Explain how the Pinball Loss function optimizes a 95th percentile (Q95) model.

**Answer:** Pinball loss penalizes under-predictions by  $\alpha = 0.95$  and over-predictions by  $1 - \alpha = 0.05$ . Because under-estimating water level is penalized 19x more heavily ( $0.95 / 0.05$ ), the model shifts its predictions upward until exactly 95% of observations fall below the curve.

### Q12: What role does Alembic play in database lifecycle management?

**Answer:** Alembic tracks SQLAlchemy ORM schema modifications and generates version-controlled SQL migrations. This allowed adding the ML prediction and model registry tables (migration 310cc77dbeeae) to live PostgreSQL without manual table drops or data loss.

**Q13: What happens if the ESP32 loses Wi-Fi connectivity during a flood?**

**Answer:** The firmware enters an offline mode, continuing physical sensor sampling and flashing an error LED. When Wi-Fi reconnects, it resumes HTTP POST transmission automatically without requiring a manual hardware reset.

**Q14: Why is WebSocket protocol used instead of HTTP polling for live dashboard updates?**

**Answer:** HTTP polling sends repetitive GET requests every second, wasting server CPU and bandwidth. WebSockets establish a single persistent, full-duplex TCP socket, allowing the backend to broadcast live JSON telemetry frames to browsers in under 1.5 milliseconds.

**Q15: How are cyclical temporal features (hour_sin, hour_cos) encoded?**

**Answer:** Linear hour numbers (0 to 23) create an artificial discontinuity between 23:00 and 00:00. Mapping hours onto a circle using  $\sin(2\pi \cdot h/24)$  and  $\cos(2\pi \cdot h/24)$  ensures 23:00 and 00:00 are adjacent in Euclidean space, enabling trees to learn continuous diurnal evapotranspiration.

**Q16: Why was capacitive soil moisture chosen over resistive probes?**

**Answer:** Resistive probes expose bare copper traces to DC current, causing rapid galvanic corrosion within weeks. Capacitive probes insulate electrodes behind solder mask, measuring soil dielectric capacitance without metallic contact, ensuring long-term field survivability.

**Q17: What is the physical limitation of ultrasonic distance sensors?**

**Answer:** Ultrasonic sensors possess an acoustic deadband (blind zone) of 20-25 cm where echoes return before the receiver circuitry can switch. Furthermore, speed of sound varies with temperature, requiring compensation via  $v = 331.3 + 0.606 \cdot T$ .

**Q18: What is non-crossing monotonicity enforcement in quantile regression?**

**Answer:** Because Q05, point, and Q95 models train independently, rare inputs can cause quantile crossing ( $q05 > \text{point}$ ). FloodWatch enforces monotonicity in `predictor.py` via  $q05 = \min(q05, \text{point})$  and  $q95 = \max(q95, \text{point})$ , guaranteeing physical validity under all conditions.

**Q19: List the 5 weighted factors of the FloodWatch Risk Engine.**

**Answer:** Water Level vs Threshold (35%), Water Surge Rise Rate (25%), Current Rainfall Rate (15%), Catchment Soil Moisture (15%), and 24-Hour Rainfall Forecast (10%).

**Q20: What security measures protect the telemetry endpoint?**

**Answer:** The `/api/v1/telemetry` endpoint requires pre-shared API key authentication via the `X-API-Key` HTTP header, rejecting unauthorized network traffic. In production, TLS 1.3 encryption prevents packet sniffing.

**Q21: What is the Manning formula and how was it used in research?**

**Answer:** Manning's formula ( $Q = 1/n \cdot A \cdot R^{2/3} \cdot S^{1/2}$ ) relates open-channel flow to geometry, slope, roughness, and depth. In Phase 4A, it was inverted to derive realistic river stage (m) from GloFAS discharge reanalysis data ( $\text{m}^3/\text{s}$ ).

**Q22: Why was Chart.js chosen over D3.js for the frontend?**

**Answer:** Chart.js renders directly to HTML5 canvas rather than thousands of SVG DOM nodes, offering vastly superior rendering performance for streaming time series. It natively supports gradient fills and threshold plugin lines with minimal bundle size.

**Q23: How does FloodWatch calculate calibrated flood probability?**

**Answer:** It estimates forecast standard deviation from the empirical 90% quantile corridor width ( $\sigma = [q95 - q05] / 3.29$ ) and evaluates the Gaussian cumulative distribution tail relative to the 3.0m Warning Threshold:  $P = (1 - \Phi(z)) \cdot 100\%$ .

**Q24: What is the purpose of the alert_history table in PostgreSQL?**

**Answer:** Whenever an alert is created, acknowledged, or resolved, the action is logged to `alert_history` with user metadata and timestamps. This creates an unalterable, legally defensible audit trail for civil defense reviews.

**Q25: Can this prototype be deployed in any river basin globally?**

**Answer:** Yes. The software, database, and hardware designs are catchment-agnostic. Deploying to another river only requires updating station coordinates, configuring local warning/danger thresholds, and retraining XGBoost with local meteorological and discharge records.

**Q26: What is the sampling frequency of the ESP32 station?**

**Answer:** Standard telemetry transmission occurs every 30 seconds. During rapid rise rate events ( $dh/dt \geq 0.10$  m/hr), the firmware dynamically accelerates sampling to 10-second intervals to capture surge dynamics.

**Q27: How does FloodWatch handle multiple sensors reporting concurrently?**

**Answer:** FastAPI handles concurrent requests asynchronously using non-blocking worker threads. PostgreSQL manages parallel writes via connection pooling and ACID row-level locking.

**Q28: What is an Antecedent Soil Saturation index?**

**Answer:** It represents the percentage of available soil pore space filled with water. Saturated soil has zero infiltration capacity, converting 100% of subsequent rainfall into immediate surface runoff.

**Q29: What is the purpose of the status LED on the ESP32?**

**Answer:** Blinking at 2Hz indicates Wi-Fi searching. Solid light indicates successful Wi-Fi connection and active telemetry transmission. Rapid strobe indicates a hardware sensor read failure.

**Q30: What is the ultimate goal of the FloodWatch platform?**

**Answer:** To provide vulnerable valley communities with an autonomous, resilient, sub-second flood warning system that combines edge IoT sensing with predictive AI while guaranteeing that deterministic physical safety rules remain supreme.

SECTION 19 — CURRENT PROJECT STATUS

The current implementation state of the FloodWatch project is audited below, clearly separating completed and verified software from future hardware tasks:

Project Architecture Module	Implementation Status	Current Verified Operational State	Next Required Engineering Action
Software Architecture	COMPLETED	End-to-end event-driven microservice pipeline fully integrated	Maintain active production services
React Dashboard	COMPLETED	Production Vite bundle compiled in 46.99s with 0 errors	Connect live hardware telemetry stream
FastAPI Backend	COMPLETED	REST APIs & WebSockets active; 9/9 automated test suites passed	Maintain backend service on port 8000
PostgreSQL 16 Engine	COMPLETED	Native engine on port 5432 with 13 tables; restart test passed	Maintain periodic ACID transaction backups
Alembic Migrations	COMPLETED	Revisions e9f898701da2 and 310cc77dbeae applied to head	Version future schema modifications
Rule-Based Risk Engine	COMPLETED	Deterministic 5-factor weights (35/25/15/15/10) verified	Calibrate prototype physical thresholds
Alert Engine & History	COMPLETED	Multi-threshold breach detection & audit trail logging verified	Configure SMS / email emergency dispatch
WebSocket Dispatcher	COMPLETED	Sub-1.5ms non-blocking real-time broadcasting verified	Maintain active browser socket connections
Simulator Engine	COMPLETED	Normal, Heavy Storm, and Flash Flood scenarios functional	Demonstrate during project evaluation viva
Historical ML Research	COMPLETED	26,280 hourly records (2021-2023) curated and validated	Retrain with local river data upon field deployment
ML Production Integration	COMPLETED	9 direct XGBoost models loaded; $R^2 > 0.97$ ; Q05/Q95 corridors verified	Maintain advisory status and monitoring
Physical Hardware Sensors	PENDING	Bill of Materials procurement specification completed	Procure ESP32, JSN-SR04T, soil probe, and rain gauge
ESP32 Integration	PENDING	C++ firmware completed in <code>iot/esp32_firmware/</code> directory	Flash sketch via Arduino IDE and test Serial output
Sensor Integration	PENDING	Electrical wiring schematic & voltage divider design completed	Assemble breadboard circuit with 1k $\Omega$ /2k $\Omega$ divider
Physical Prototype Box	PENDING	CAD blueprint and wet/dry zone layout designed	Assemble acrylic container, gantry, and pump

## SECTION 20 — FINAL PROJECT SUMMARY

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The **FloodWatch** platform represents a complete, verified, and scientifically defensible cyber-physical flood monitoring and early warning system. By unifying eight foundational engineering pillars—Internet of Things edge transduction, sub-second real-time monitoring, asynchronous backend microservices, enterprise relational persistence, deterministic hydrological risk analytics, multi-horizon machine learning, reactive web visualization, and physical hydraulic prototyping—FloodWatch bridges the critical gap between environmental sensing and life safety.

### Key Engineering Achievements:

1. **Deterministic Safety Primacy:** Established an inviolable safety hierarchy where physical threshold breaches trigger immediate alarms, proving that machine learning remains strictly advisory and cannot override physical safety rules.
2. **Statistically Valid Quantile Prediction:** Rejected arbitrary heuristic multipliers in favor of dedicated pinball-loss XGBoost regressors delivering calibrated 90% empirical uncertainty corridors ([Q05, Q95]) with enforced non-crossing monotonicity.
3. **Enterprise Relational Integrity:** Deployed PostgreSQL 16 with thirteen fully migrated, foreign-key-constrained tables, maintaining unalterable alert history audit logs for civil protection compliance.
4. **Sub-Second Operator Reactivity:** Achieved sub-1.5ms WebSocket dispatch latency, enabling operators to observe hydraulic surges and rate-of-rise ( $dh/dt$ ) spikes dynamically without page refreshes.
5. **Hardware Prototype Blueprint:** Formulated exhaustive component procurement specifications, electrical wiring diagrams with voltage divider safety calculations, sensor calibration formulas, and a 16-step implementation sequence for the physical demonstrator.

FloodWatch stands as an exemplary, publication-grade engineering achievement that bridges Internet of Things telemetry, environmental hydraulics, data science, and civil protection to preserve human lives and urban infrastructure.